A GUI for Magboltz

IFIN-HH (Romania)

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WG4 — Modeling and Simulations

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Magboltz: Context & Workflow

Why it matters

- Monte Carlo electron transport simulation
- Fundamental tool for detector physics
- Used across MPGDs, TPCs, RPC studies

CLI friction

- Text-based input cards
- Manual output parsing
- Iterative tuning & plotting scripts
- Student onboarding difficulty

Cornerstone physics engine, workflow-heavy interface.

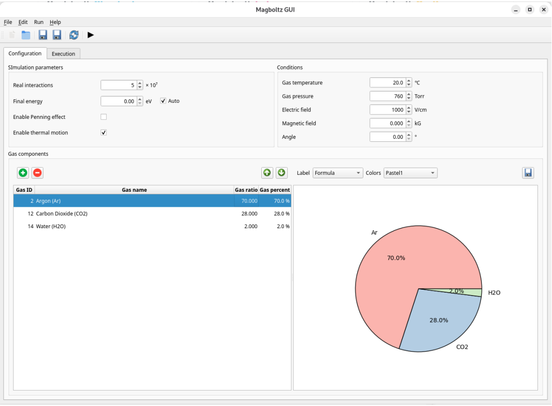
Workflow Pipeline

Input $\rightarrow$ Run $\rightarrow$ Parse $\rightarrow$ Export $\rightarrow$ Plot

We change the workflow, not the physics.

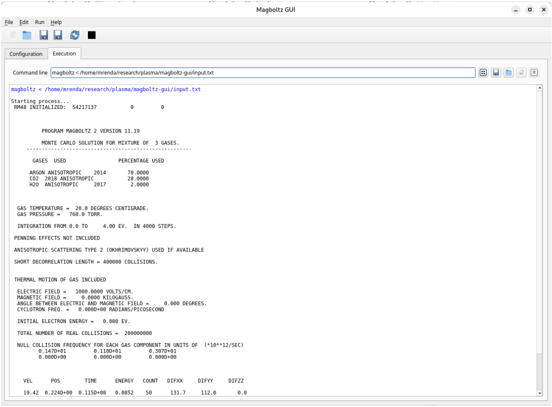
Magboltz GUI

- Python + PyQt6 application
- Cross-platform
- Designed as a workflow layer, not a replacement
- Core functions:
  - Input file creation & editing
  - Gas mixture editor & visual pie chart
  - Magboltz execution wrapper
  - Output parsing → structured model
  - Data export → CSV / JSON / XML
  - Built-in plotting tools



Execution Workflow

- Direct Magboltz invocation
- Command line transparency
- Live stdout/stderr streaming
- Run/Stop controls
- Raw output persistence
- Reopen previous runs



What This Does NOT Do

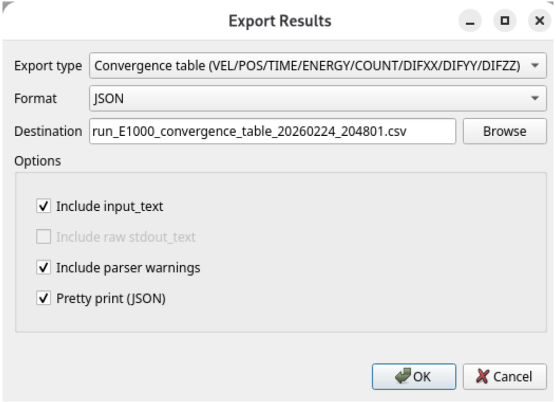
- Does **not** change Magboltz physics, models, or algorithms
- Does **not** alter input semantics or units
- Does **not** replace the CLI (it wraps it)
- Does **not** ship the Magboltz binary

Validation & Trust

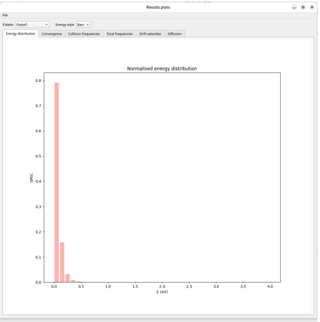
- Parsed outputs are **1:1** with Magboltz text output
- No unit conversions; values are preserved as printed
- Export schema is versioned (`v1.0`)
- Raw output & parser warnings are retained

From Text Output → Structured Data

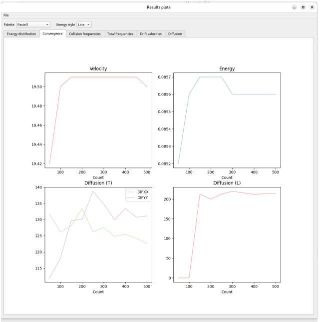
- Canonical RunResult model
- Extracted:
  - Transport coefficients
  - Collision frequencies
  - Convergence tables
  - Energy distributions
- Export formats:
  - CSV
  - JSON
  - XML
- Reproducible analysis pipelines



Built-in Plotting Tools



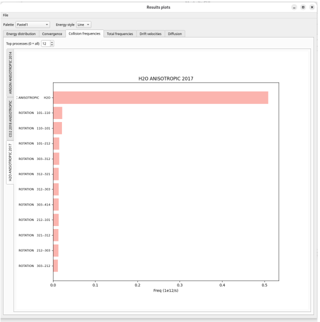
Energy distribution



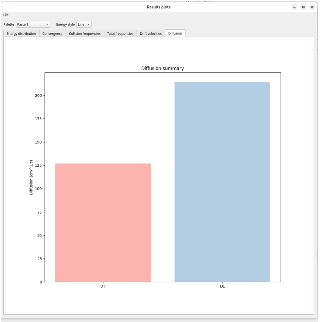
Convergence

Immediate visual feedback without external scripts.

Built-in Plotting Tools (cont.)



Collision frequencies



Diffusion summary

Immediate visual feedback without external scripts.

Availability & License

Project repository

- GitLab: `https://gitlab.com/micrenda/magboltz-gui/`
- License: MIT
- Documentation: README in the repository

Setup: pip (bundled Qt)

Recommended when you want Qt bundled with the app.

- Python environment
- Magboltz binary available in PATH

Commands:

```
pip install "magboltz-gui[qt]"
magboltz-gui
```

Setup: pip (system Qt)

Recommended on Linux when using system PyQt6.

- Install system Qt bindings
- Magboltz binary available in PATH

Commands (Debian/Ubuntu):

```
sudo apt install python3-pyqt6
pip install magboltz-gui
magboltz-gui
```

Limitations & Next Steps

- No batch sweeps yet; single-run workflow
- Limited automation for parameter scans
- Next: parameter scan runner + input templates
- Next: headless export-only mode for pipelines

In Memoriam

Stephen F. Biagi

(1949–2025)

We dedicate this work to his memory.

His pioneering contributions to modeling electron transport in gases profoundly shaped detector physics.

This interface is a modest tribute to his legacy.